

SYLLABUS :-

ME20001: Dynamics (3-1-0-4) Kinematics of particles: representation of motion of particles in various coordinate systems, relative motion in translating frames, constrained motion; Kinetics of particles: Newton's laws of motion, equations of motion for unconstrained and constrained motion, work-energy relation, conservation of energy, conservative and non-conservative forces, potential energy, impulse-momentum relation, angular momentum, conservation of momentum, applications; Kinetics of a system of particles: Newton's second law, work-energy relation, impulse-momentum relations, conservation laws, steady and variable mass flow systems; Plane kinematics of rigid bodies: Kinematics of rigid bodies, instantaneous center of rotation, kinematics in rotating frames and relative motion; Plane kinetics of rigid bodies: Linear and angular momentum, equations of motion, work-energy relation, impulse-momentum relation, conservation laws; Introduction to spatial dynamics of rigid bodies: Kinematics in rotating frames and relative motion, angular momentum, kinetic energy, equations of motion, special cases of parallel-plane motion, motion about a fixed point and gyroscopic motion Book: x Engineering Mechanics: Dynamics, J.L. Meriam and L.G. Kraige x Vector Mechanics for Engineers, F.P. Beer et al.